

Robots in Context

Programming Exercise 6.2

Leon Bodenhagen lebo@mmmi.sdu.dk

Associate Professor

SDU Robotics, Maersk Mc-Kinney Moller Institute



Programming Exercise - Task

Implement PRM for a 2D planning problem.

- You can assume the robot to be a point to reduce the complexity.
- Use for instance the same map as before, e.g., for Bug.

Requirements:

1. Verify that the algorithm works for a number of cases (you can use the same map and build different roadmaps and use the same set of start/goals, e.g. 5)
2. Report e.g. success, path length and other aspects you find relevant
3. Discuss some of your choices, e.g. sampling strategy or connection strategy and investigate the impact (might be qualitative)
4. Optional: check for 'difficult' nodes and extend the roadmap 

